

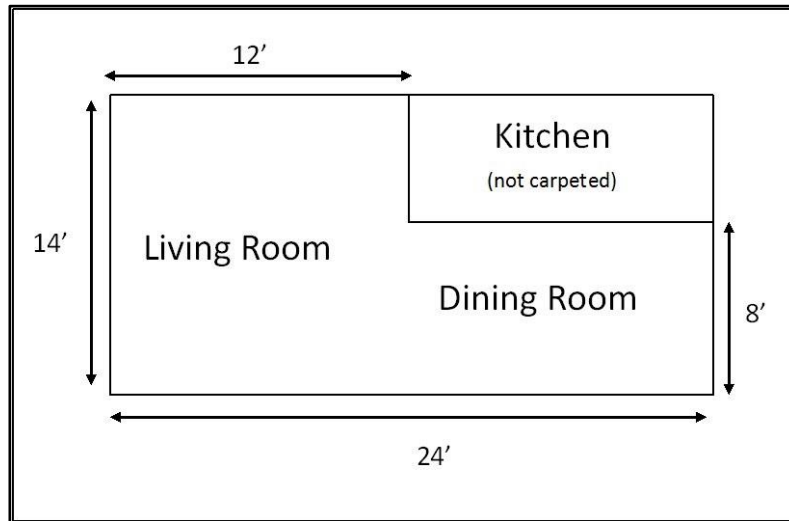
MATH SKILLS PREP – SAMPLE QUESTIONS

Here are some sample math questions to get you prepared for the math portion of your exam. Note that these are similar to the type of questions you will be asked on the exam, but of course the actual exam questions will vary. The best way to practice this is to use scratch paper and a calculator, just as you will have at the exam. You will find the answers at the end.

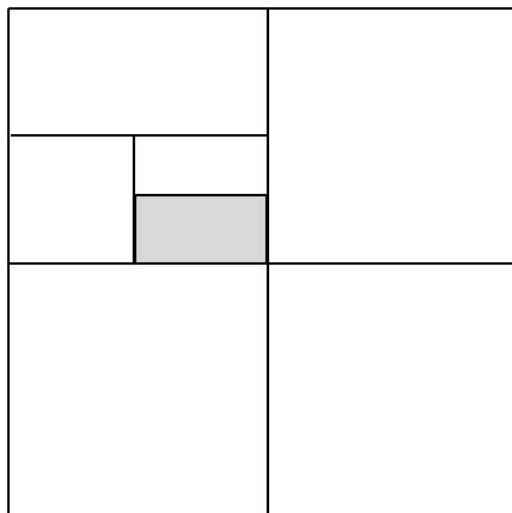
1. Your client, Jill, offers to pay you a 5% commission on the sale of her home, which you have listed at \$260,000. If the house sells for its list price, and your share of the commission is 2.5%, what will your commission be?
 - A. \$13,000
 - B. \$1,300
 - C. \$650
 - D. \$6,500
2. Your client's property is 67,850 square feet. Approximately how many acres is the property?
 - A. 30
 - B. 1.5
 - C. 15.5
 - D. 3
3. Your client's home sold for \$385,000. Brokerage fees are 6%, closing costs are \$7,380, and the loan payoff balance is \$210,000. What will your client's net proceeds be from the sale?
 - A. \$144,520
 - B. \$175,000
 - C. \$210,000
 - D. \$138,750
4. Sherman invested \$150,000 in an apartment building. The first year he received \$20,000 in pre-tax cash flow. What is his cash-on-cash return?
 - A. 13.33%
 - B. \$20,000
 - C. 12.7%
 - D. -\$130,000
5. Your client purchased a property for \$300,000. Net operating income is \$15,000. What is the cap rate for the property?
 - A. 8%
 - B. 15%
 - C. 6%
 - D. 5%
6. As a buyer's agent, you earn 2.5% on the sale of a home. This is one half of total commission paid. If the total commission paid is \$5,400, what was the sales price of the home?
 - A. \$210,000
 - B. \$108,000
 - C. \$128,000
 - D. \$185,640

7. Sales agent Timothy has a 60/40 split with his broker (with Timothy getting 60). He just sold a home for \$395,000 which paid a 3% commission to Timothy's firm. How much will Timothy receive?
- A. \$3,800
 - B. \$4,740
 - C. \$11,850
 - D. \$7,110
8. Your client sold her home for \$290,000. She paid out 4% in broker fees and another .75% in closing costs. If her loan payoff is \$170,000, what will her net proceeds be?
- A. \$120,000
 - B. \$113,775
 - C. \$106,225
 - D. \$112,750
9. Your clients are looking for a property that is at least 3 acres but they don't want anything larger than 5 acres. Which of the following would meet their criteria?
- A. A property that is 120 feet by 740 feet
 - B. A property that is 158,000 square feet
 - C. A property that is 105 feet by 110 feet
 - D. A property that is 250,850 square feet
10. Comparable homes in your area are selling, on average, for 3% under list price. If your clients have their house listed for \$175,000, what might they expect to sell their house for?
- A. \$169,750
 - B. \$157,850
 - C. \$172,000
 - D. \$168,900
11. Cameron has an investment property for which he paid \$110,000. It has a cap rate of 5%. What is his net operating income per month?
- A. \$5,000
 - B. \$5,528
 - C. \$612.35
 - D. \$458.33
12. Bijou is purchasing a townhome for \$270,000. She has put \$5,000 down as earnest money, and is getting an 80/20 loan. Her closing costs are \$5,125. How much will she need to bring to closing to cover her down payment and closing costs?
- A. \$51,250
 - B. \$54,125
 - C. \$59,125
 - D. \$58,250
13. Your seller, Steve, owes \$115,000 on his house. Assuming Steve's closing costs will be 6.5% of the sales price, at what price must his house sell for to break even?
- A. \$107,525
 - B. \$109,375
 - C. \$122,995
 - D. \$115,000
14. What amount of interest does Acme Bank receive at the end of 2 years on a \$540,000 loan at 5.75% interest?
- A. \$31,500
 - B. \$62,100
 - C. \$57,525
 - D. \$28,750

15. What amount does an investment return after 2 months if its annual rate of return is 12% and the initial investment is \$200,000?
- \$6,000
 - \$12,000
 - \$2,000
 - \$4,000
16. Your clients, the Mathers, have made an offer on a house, but the carpeting needs to be replaced in the living room and dining room. If the sellers are willing to give the Mathers a \$3,500 allowance for the carpeting, what is the highest price per square foot they can pay for their new carpeting?



- \$12
 - \$13
 - \$18
 - \$23
17. You see a legal description that reads: S $\frac{1}{2}$ SE $\frac{1}{4}$ NW $\frac{1}{4}$ Section 10, Township Territorial. How large is that parcel in acres?
- 40 acres
 - 10 acres
 - 25 acres
 - 20 acres



Answer Key

1. D. [$\$260,000 \times .025 = \$6,500$]
2. B [There are 43,560 square feet in an acre. $67,850 / 43,560 = 1.56$]
3. A [$\$385,000 \times .06 = \$23,100$. $\$23,100 + \$7,380 + \$210,000 = \$240,480$. $\$385,000 - \$240,480 = \$144,520$]
4. A [$\$20,000 / \$150,000 = .1333$ (13.33%)]
5. D [Cap rate = Income/Value. Income is \$15,000. Value is \$300,000. $15,000 / 300,000 = .05$ (5%)]
6. B [Total commission = $2 \times \$2,700 = \$5,400$. $\$5400 / .05 = \$108,000$]
7. D [Commission = 3%. $\$395,000 \times .03 = \$11,850$. $\$11,850 \times .6 = \$7,110$]
8. C [Commission + closing costs = 4.75%. $\$290,000 \times .0475 = \$13,775$. $\$13,775 + \$170,000 = \$183,775$. $\$290,000 - \$183,775 = \$106,225$]
9. B [$158,000 / 43,560 = 3.63$ acres]
10. A [$\$175,000 \times .97 = \$169,750$]
11. D [$\$110,000 \times .05 = \$5,500 / 12$ months = \$458.33]
12. B [$\$270,000 \times .20 = \$54,000 + \$5,125 = \$59,125 - \$5,000 = \$54,125$]
13. C [$\$115,000 / .935$ (100% - 6.5% = 93.5% or .935) = \$122,995]
14. B [$\$540,000 \times .0575 = \$31,050 \times 2 = \$62,100$]
15. D [$200,000 \times .12 = \$24,000 / 12 = \$2,000 \times 2 = \$4,000$]
16. B [$(8 \times 12 + 14 \times 12 = 22 \times 12$ or 264SF. $\$3500 / 264 = \13.26]
17. D [The easy way to compute this is to multiply the fractions across: S $\frac{1}{2}$ SE $\frac{1}{4}$ NW $\frac{1}{4} = 1/32$. A section is one square mile, or 640 acres. This is $1/32$ of 640 acres, or $640 / 32 = 20$ acres]

